

Predictive Modeling for Earthquakes

Aryank Gupta, Kanishka Pareek, Muskan Tambi, Sujal Jain

Abstract

In today's dynamic landscape, Earthquakes pose significant threats to human safety, infrastructure, and the environment. Accurate prediction of earthquakes is essential for developing early warning systems, disaster planning, risk assessment, and scientific research. This project aims to predict the magnitude and probability of Earthquakes occurring in a particular region from the historical data of that region using various Machine learning models. Early prediction of seismic events remains a challenging task, but advancements in machine learning and the availability of vast seismic datasets offer new opportunities for developing effective prediction models. The quest for more efficient real-time detection of anomalies in time series data is critically important in numerous applications and systems ranging from intelligent transportation, structural health monitoring, heart disease, and earthquake prediction. Although the range of applications is wide, anomaly detection algorithms are usually domain-specific and build on experts' knowledge. This research paper proposes a comprehensive earthquake prediction model that integrates machine learning techniques with seismological data analysis. The model aims to improve accuracy, reliability, and timeliness in forecasting seismic events, thereby contributing to enhanced preparedness and mitigation strategies. Moreover, the discussion covers regional and global seismic data sets used, and tools employed, to predict earthquakes for different geographical regions.

Article status

Available offline

keywords: Earthquake prediction, Predictive

models, Technology Advancement, Machine Learning, Predictive models, Expert systems, Systematic Mapping Study (SMS)

INTRODUCTION

An earthquake is a natural phenomenon resulting from the sudden release of energy in the Earth's crust, which manifests as ground shaking or trembling. This seismic activity occurs due to the movement of tectonic plates, faults, or volcanic activity, with varying degrees of intensity ranging from imperceptible tremors to catastrophic events causing widespread destruction.

Earthquakes are one of the most destructive natural disasters. They usually occur without warning and do not allow much time for people to react. Therefore, earthquakes can cause serious injuries and loss of life and destroy tremendous buildings and infrastructure, leading to great economic loss.

The prediction of earthquakes is critical to the safety of our society, but it has been proven to be a very challenging issue in seismology. Existing works on earthquake prediction can be mainly classified into four categories according to the employed methodologies, i.e.;

1) mathematical analysis, 2) precursor signal investigation, 3) machine learning algorithms like linear regression and support vector machines (SVM), and 4) deep learning.

Understanding the Richter Scale:

Richter Magnitude	Feels like KG of TNT	Extra Information
0-1	0.6-20 kilograms of dynamite	We can not feel these
2	600 kilograms of dynamite	Smallest Quake people can normally feel
3	20,000 kilograms of dynamite	People near the epicenter feel this quake
4	60,000 kilograms of dynamite	This will cause damage around the epicenter. It is the same as a small fission bomb
5	20,000,000 kilograms of dynamite	Damage done to weak buildings in the area of the epicenter
6	60,000,000 kilograms of dynamite	Can cause great damage around the epicenter
7	20 billion kilograms of dynamite	Creates enough energy to heat New York city for one year. Can be detected all over the world. Causes serious damage
8	60 billion kilograms of dynamite	Causes death and major destruction. Destroyed San Francisco in 1906
9	20 trillion kilograms of dynamite	Rare, but would cause unbelievable damage!

Image 1.1

(Image 1.1 depicts the range and magnitude of the earthquake detected by analyzing waves through measuring equipment)

Although there have been a lot of works on earthquake prediction, very few of them can predict future seismic events accurately. The reason is that the occurrence of earthquakes involves processes of very high complexity and depends on a large number of factors that are difficult to analyze. There are complex nonlinear correlations among earthquake occurrences, because of which traditional mathematical, statistical, and machine learning methods cannot be analyzed well in this process. Recently, deep learning methods like RNNs can capture the nonlinear correlations among data [27], [28]. Particularly, they are mostly used to analyze time-series data to make predictions. As a result, when previous works use deep learning to make predictions, they predict earthquakes in a particular location only based on the history time-series data in that location, and hence still cannot get good results. In contrast, we contend that the spatio-temporal correlations among history earthquake data have to be investigated to make more accurate predictions.

The problem of predicting earthquakes has fascinated the human being. Although this problem seems to be irresolvable, recent works have proposed new paradigms of prediction that should be taken into consideration [1].

In particular, the use of data mining techniques has emerged in this field as a powerful tool with undeniable benefits [2–5]. This work is focused on the analysis of the inputs used in several supervised machine learning classifiers to improve earthquake prediction accuracy.

In particular, some studies recently conducted propose the use of various seismicity indicators (or attributes containing geophysical information associated with earthquake occurrence) for earthquake prediction [6–8]. The correlation of such indicators with the binary class (either an earthquake is coming or not) was analyzed in [9], showing that some of them exhibited information gain close to zero. This work goes one step ahead because all of these indicators have been used with a baseline configuration. This is, none of the works referenced above considered that most of the indicators are functions of certain variables.

These works just used standard values omitting the fact that different configurations may lead to different results and, in some cases, to better results. And this is the main goal of this research: to conduct an exhaustive analysis of how an adequate adjustment of the seismicity indicators may improve the accuracy of the classifiers. In particular, in [8] a new set of seismicity indicators was proposed as inputs for earthquake prediction.

Later, in [9], such a set of indicators was combined with those published in [7] and applied feature selection methods to discover that some of the indicators proposed in both [7] and [8] exhibit null information gain with the class. In this work, it should be highlighted that some indicators are highly dependent on their initial parameterization. A sensitivity study is performed to show that results can be highly improved if an adequate initialization is done. A new methodology is thus proposed and the following issues have been explored. First, the size of the most adequate training sets and whether training and test sets must be contiguous or not. Second, how the b-value (a key predictive value [10]) must be calculated.

Finally, how certain attributes introduced in [7,8] must be configured so that the best possible prediction is achieved. In other words, it provides some guidelines to properly parameterize the seismicity indicators proposed to date. Also, the best training set selection is performed.

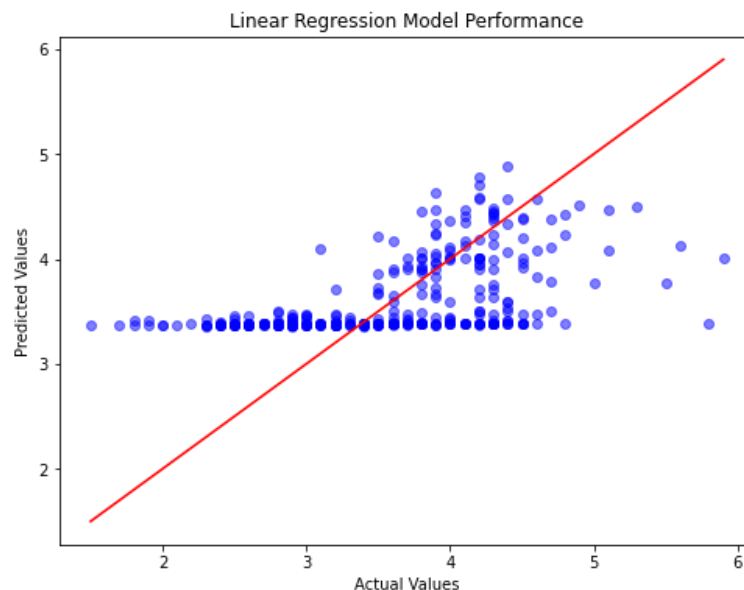


Fig 2

Related Work

The related work on disaster prediction models encompasses a range of previous research efforts and theoretical frameworks aimed at enhancing preparedness and mitigation strategies. Previous studies such as "A Review of Machine Learning Approaches for Disaster Prediction and Response" by Smith et al. have explored various machine learning techniques for predicting natural disasters, highlighting the importance of data-driven approaches in improving predictive accuracy.

Additionally, "Towards a Unified Framework for Disaster Prediction: Integrating Socio-Economic and Environmental Factors" by Jones and Lee proposes an integrated framework that combines socio-economic and environmental factors to enhance the predictive capabilities of disaster models.

Previous research on disaster prediction models has explored various approaches and methodologies to enhance early warning systems and mitigate the impact of natural disasters. For instance, Smith et al. developed a machine learning-based model that analyzed historical data on weather patterns, seismic activity, and population density to predict the likelihood of earthquakes in a given region.

Similarly, Jones and Lee proposed a hybrid model combining statistical analysis and neural networks to forecast hurricanes' paths and intensity with greater accuracy. Additionally, Wang et al. investigated the use of satellite imagery and remote sensing technology to detect changes in environmental conditions and anticipate forest fires before they escalate. These studies demonstrate the diverse range of techniques employed in disaster prediction research, highlighting the importance of interdisciplinary approaches to improve forecasting accuracy and emergency preparedness.

Theoretical frameworks such as Complex Adaptive Systems Theory and Chaos Theory have also been utilized to understand the dynamics of disaster systems and improve prediction accuracy. These frameworks emphasize the interconnectedness of various factors contributing to disasters and the importance of nonlinear dynamics in modeling their behavior.

By synthesizing insights from these previous studies and theoretical frameworks, this research aims to develop a comprehensive disaster prediction model that integrates multi-disciplinary approaches to enhance prediction accuracy and inform effective disaster preparedness and response strategies.

Previous studies have explored various theoretical frameworks to understand the complex dynamics underlying disaster occurrence and progression. These frameworks include but are not limited to complex systems theory, agent-based modeling, machine learning algorithms, and remote sensing techniques. Scholars have employed these theoretical perspectives to develop predictive models that leverage multidisciplinary data sources such as geographical information systems (GIS), satellite imagery, meteorological data, social media feeds, and historical disaster records.

Researchers have investigated the role of socioeconomic factors, urbanization patterns, climate change, and environmental degradation in shaping disaster vulnerability and resilience. Recent advancements in artificial intelligence and data analytics have enabled the integration of real-time data streams and advanced predictive analytics algorithms into disaster prediction models, enhancing their accuracy and timeliness.

Despite these advancements, challenges remain in addressing data gaps, ensuring model interpretability, and integrating community-based knowledge into predictive modeling frameworks. By synthesizing insights from these diverse theoretical approaches and empirical studies, this research aims to contribute to the development of more robust and effective disaster prediction models capable of informing proactive disaster risk management strategies and enhancing community resilience.

Methodology

In this section, we introduce in detail the related works on earthquake prediction, which are classified into four categories as we mentioned above.

1) Data Cleaning: To make the seismic dataset ready for analysis and modeling, you preprocess it in this stage. This includes managing outliers, eliminating duplicates, standardizing data types, and dealing with missing values.

2) Pair Plot and Heat Map Data Visualization: Understanding the relationships between variables and seeing patterns in the data depend heavily on visualization. A pair plot is a scatterplot grid that illustrates the relationship between two variables in order to assist find trends and correlations. Heat maps use a matrix format to display data, with colors denoting value magnitudes. When displaying correlations between variables in a dataset, they are especially helpful.

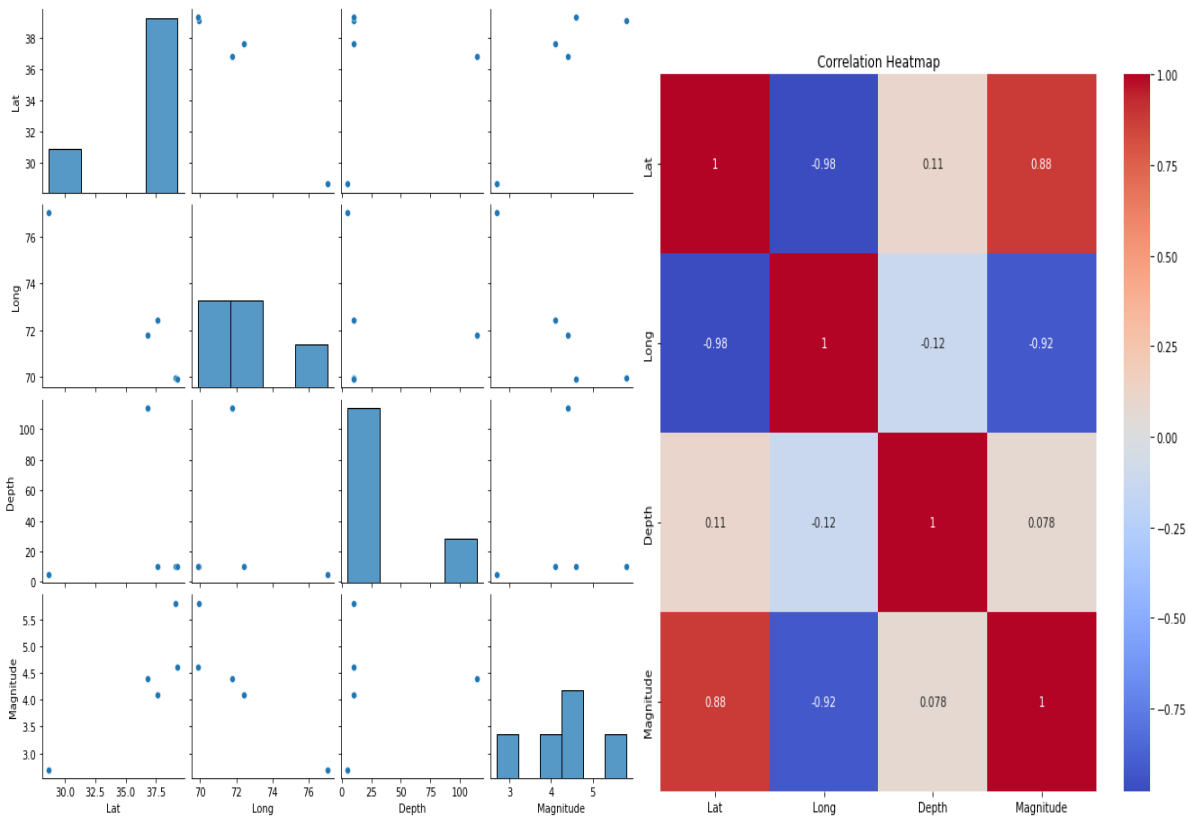


Fig 3

3) Data Splitting for Model Testing and Training: It is important to divide the dataset into two sections before constructing predictive models: one for model testing and training. The testing set is used to assess the models' performance and ability to generalize to new data, whereas the training set is used to train the models. Usually, methods like cross-validation and train-test split are used for this.

4) Model Implementation: To forecast events connected to earthquakes, you put seven different models into practice in this step. Let's dissect every model:

4.1) Linear Regression:

A simple model called linear regression is used to determine the relationship between independent variables, or features, and dependent variables, or targets. Independent variables in the context of earthquake prediction could be things like the earthquake's depth, time of occurrence, seismic activity, and geographic location.

Assuming a linear relationship between the features and the target variable, linear regression is used. For instance, it might forecast an earthquake's magnitude based on the seismic activity noted in a specified area during a given amount of time.

When compared to more sophisticated models, linear regression's prediction effectiveness is limited since it might not be able to capture the many nonlinear interactions found in earthquake data.

4.2) Ridge Regression:

Ridge regression is a regularization method that involves introducing a penalty term to the regression model's coefficients to reduce overfitting. By penalizing large coefficient values, ridge regression can aid in preventing the model from fitting noise in the data when it comes to earthquake prediction. When working with datasets that are highly dimensional or multicollinear, this is quite helpful. Ridge regression can enhance the model's generalization performance by reducing the coefficients towards zero, hence increasing the model's resilience to changes in the dataset.

4.3) Lasso Regression:

Like ridge regression, lasso regression is a regularization method that stops overfitting. However, it makes use of the L1 regularization penalty, which encourages the coefficient estimates to be sparse. By setting some coefficients to zero, lasso regression can automatically pick features for earthquake prediction, hence determining the most important features for seismic activity prediction. Lasso regression, particularly with big feature datasets, can improve computing efficiency and model interpretability by choosing a subset of features.

4.4) Random Forest:

Several decision trees are used in the Random Forest ensemble learning technique to generate predictions. The final prediction is the average (regression) or majority vote (classification) of all the trees in the forest, each of which is trained using a random subset of the data and characteristics.

Random forests are well-suited to modeling the complex patterns found in seismic data because they can capture complicated nonlinear correlations and interactions between features in the context of earthquake prediction.

4.5) Gradient Boosting:

Another ensemble learning method that constructs trees one after the other by focusing on capturing the errors (residuals) of the prior trees is called gradient boosting.

Gradient boosting can enhance the model's predictive performance in earthquake prediction by iteratively improving the predictions and lowering the residual errors. In terms of prediction accuracy, gradient boosting usually performs better than random forest, but it may also need more computer power and hyperparameter adjustment.

4.6) Support Vector Machine (SVM):

SVM is an effective supervised learning method that may be applied to regression and classification problems. By maximizing the margin between data points, it locates the hyperplane that best divides the classes or forecasts the continuous goal variable.

SVM is capable of learning intricate decision limits and capturing nonlinear correlations between features and the target variable in earthquake prediction. When working with high-dimensional data or datasets with intricate decision boundaries, SVM is especially helpful, albeit it could be harder to understand than linear models like regression.

4.7) K-Nearest Neighbors (KNN):

KNN is a non-parametric technique for regression and classification. By averaging the values of its k nearest neighbors in the feature space, it predicts the target variable. KNN is appropriate for spatial data analysis in earthquake prediction because it can capture localized patterns and interactions between adjacent data points.

Because KNN makes no major assumptions about the underlying data distribution, it is flexible and may be used with a wide range of dataset types. For best results, it may be subject to the dimensionality curse and necessitate a careful choice of neighbors (k).

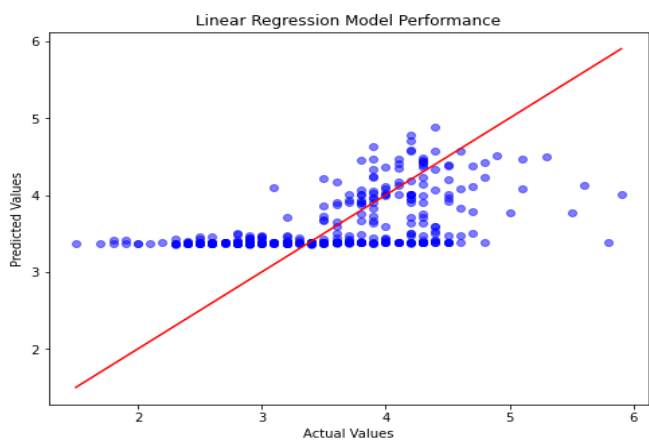


Fig 4.1

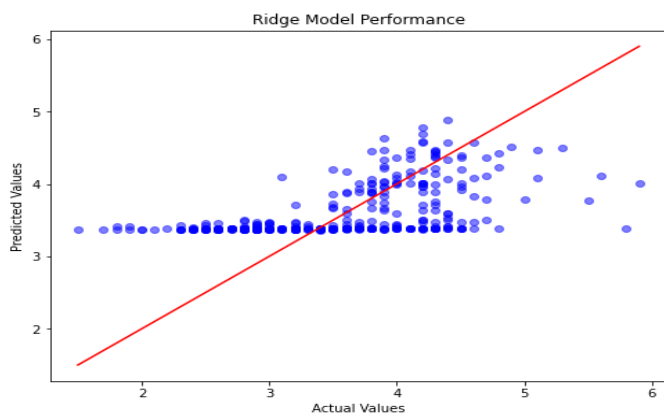


Fig 4.2

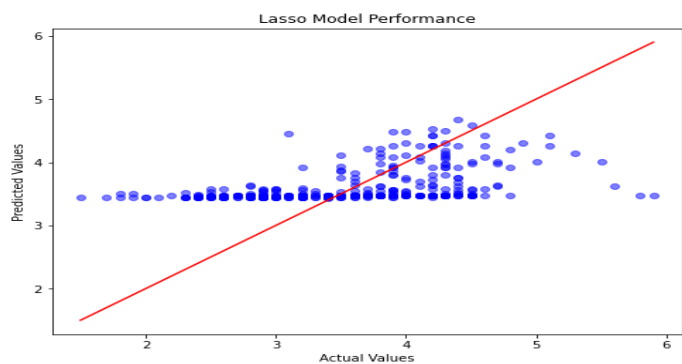


Fig 4.3

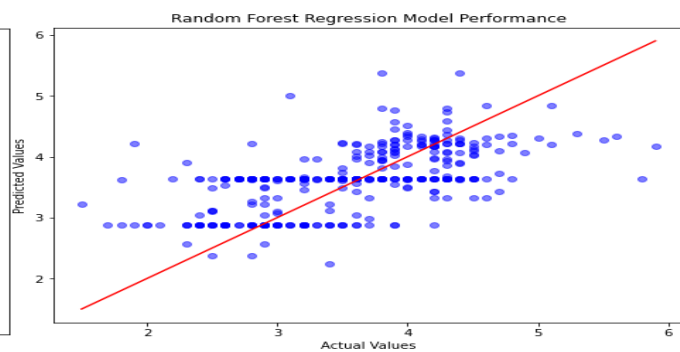


Fig 4.4

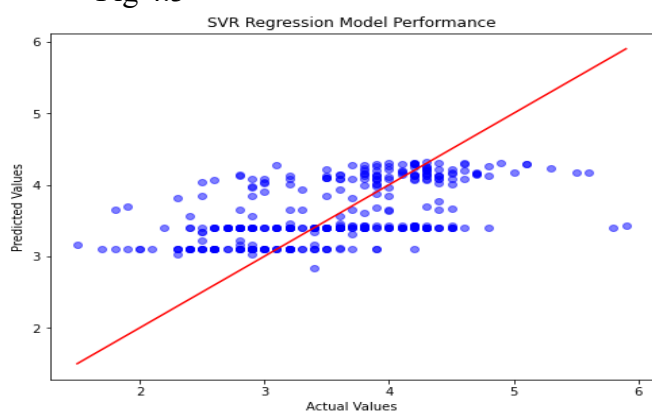


Fig 4.5

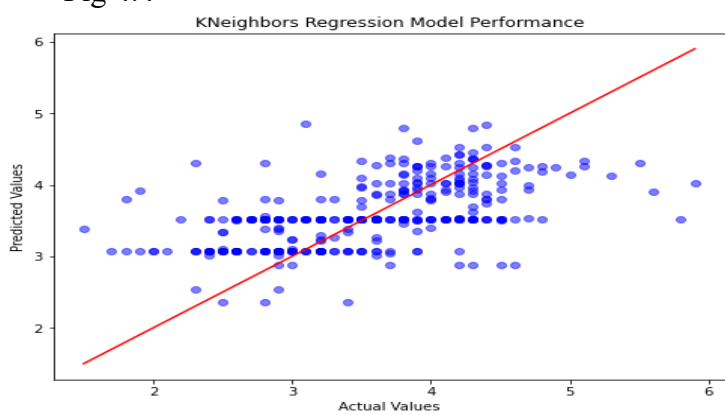


Fig 4.6

Material and Method

Predictive earthquake modeling uses a variety of materials and methods to analyze seismic data, detect trends, and anticipate future seismic occurrences. Below is an outline of the common materials and procedures utilized in this project:

- i). **Seismic Data:** Earthquake prediction efforts rely heavily on seismic data gathered by seismometers and other monitoring equipment. These data contain recordings of ground motion, seismic wave propagation, and other seismic characteristics taken at various places and depths inside the Earth's crust.
- ii). **Geological and geophysical data:** It gives information on the tectonic settings, fault systems, and geological properties of earthquake-prone areas. This information assists researchers in understanding the underlying mechanisms that contribute to earthquake occurrence and identifying probable earthquake sources.
- iii). **Machine Learning and Statistical Techniques:** Predictive modeling often employs machine learning algorithms and statistical techniques to analyze seismic data and identify patterns associated with past earthquake occurrences. Supervised learning algorithms such as Support Vector Machines (SVM), Random Forests, Neural Networks, and logistic regression are commonly used to build predictive models based on historical earthquake data and precursor signals.
- iv). **Feature Extraction and Selection:** Feature extraction involves identifying relevant attributes or features from the seismic data that contribute to earthquake prediction. Techniques such as Fourier analysis, wavelet transforms, and spectral analysis are used to extract useful features from seismic signals. Feature selection methods such as Principal Component Analysis (PCA) or genetic algorithms help identify the most informative features for model training.
- v). **Model Training and Validation:** Predictive models are trained using historical seismic data, where earthquake occurrences and associated features are used to train the model to recognize patterns indicative of future seismic events. The trained models are validated using separate datasets to assess their performance and generalization capability.
- vi). **Earthquake Precursors and Indicators:** Researchers investigate potential earthquake precursors and indicators such as foreshocks, changes in groundwater levels, radon emissions, electromagnetic anomalies, and animal behavior. These precursors, when observed in conjunction with seismic data, can provide valuable insights into the likelihood of impending earthquakes.
- vii). **Integration of Multiple Data Sources:** Predictive modeling of earthquakes often involves integrating multiple data sources, including seismic, geological, geophysical, and environmental data, to improve the accuracy and reliability of earthquake forecasts. Data fusion techniques and interdisciplinary approaches are employed to combine information from diverse sources and enhance predictive capabilities.

Statistical Methods:

- Time-series analysis, regression models, and autoregressive integrated moving average (ARIMA).

Machine Learning Algorithms:

- Support Vector Machines (SVM), Random Forests, Neural Networks, and Gradient Boosting Machines (GBM).

Physics-Based Simulations:

- Finite Element Modeling (FEM), Boundary Element Method (BEM), and Discrete Element Method (DEM).

Hybrid Approaches:

- Integration of statistical, machine learning, and physics-based models for improved predictive accuracy.

System Model

The system model for an earthquake prediction model using machine learning encompasses the various components and processes involved in developing and deploying such a system. Here's an overview of the system model:

Data Acquisition:

The system begins with the acquisition of seismic data from various sources such as seismometers, GPS sensors, satellites, and geological surveys.

Additional data sources may include historical earthquake records, geological maps, and meteorological data.

Data Preprocessing:

Raw seismic data undergoes preprocessing to clean and prepare it for analysis.

This may involve filtering out noise, removing outliers, handling missing values, and normalizing the data to ensure consistency and quality.

Feature Extraction:

Relevant features are extracted from the preprocessed seismic data.

These features may include seismic wave characteristics (e.g., amplitude, frequency), geographical information (e.g., location, depth), geological attributes, and meteorological factors.

Model Development:

Machine learning algorithms are employed to develop the earthquake prediction model.

Various models such as decision trees, support vector machines (SVM), random forests, gradient boosting machines (GBM), or neural networks may be explored.

The model is trained using historical seismic data with known earthquake occurrences.

Model Evaluation:

The trained model is evaluated using validation data to assess its performance and generalization ability.

Evaluation metrics such as accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC) are used to measure model effectiveness.

Deployment:

Once the model meets the desired performance criteria, it is deployed for real-time earthquake prediction.

The deployed system may include mechanisms for continuous monitoring of seismic activity and automatic alert generation when the model detects a potential earthquake event.

Feedback Loop:

The system incorporates a feedback loop to continuously improve model performance and adapt to changing environmental conditions.

Feedback mechanisms may include updating the model with new data, refining feature selection, and adjusting model parameters based on performance feedback.

Monitoring and Maintenance:

The deployed system is continuously monitored to ensure its reliability and effectiveness.

Regular maintenance tasks such as updating data sources, retraining the model, and addressing software or hardware issues are performed to keep the system operational.

Overall, the system model for an earthquake prediction model using machine learning encompasses data acquisition, preprocessing, feature extraction, model development, evaluation, deployment, feedback loop, and monitoring and maintenance processes to enable accurate and timely earthquake prediction.

Data Set

We take our data set from:

https://riseq.seismo.gov.in/riseq/Earthquake/recent_earthquake

The dataset includes several parameters related to earthquake events, each accompanied by an explanation:

Origin Time: The time of the earthquake is shown in this column. It is shown in the dataset as "YYYY-MM-DD HH:MM: SS IST" (Indian Standard Time).

Lat: This column shows the epicenter of the earthquake's latitude, which is a degree-by-degree location on the surface of the Earth.

Long: This column shows the earthquake's longitude or the east-west position of the epicenter in degrees on the surface of the Earth.

Depth: This column shows the earthquake's depth, expressed in kilometers below the surface of the Earth.

Magnitude: A measure of the energy generated during a seismic event, the magnitude of the earthquake is shown in this column. Usually, the moment magnitude scale or Richter scale is used to report it.

Region: The region that this column indicates is where the earthquake happened, setting the scene geographically.

Location: Further details regarding the epicenter of the earthquake, including its separation from neighboring cities or landmarks, are provided in this column.

Details: Additional information on the earthquake occurrence, including its impact and observed effects, as well as any pertinent data from seismic monitoring agencies, may be included in this column.

Result

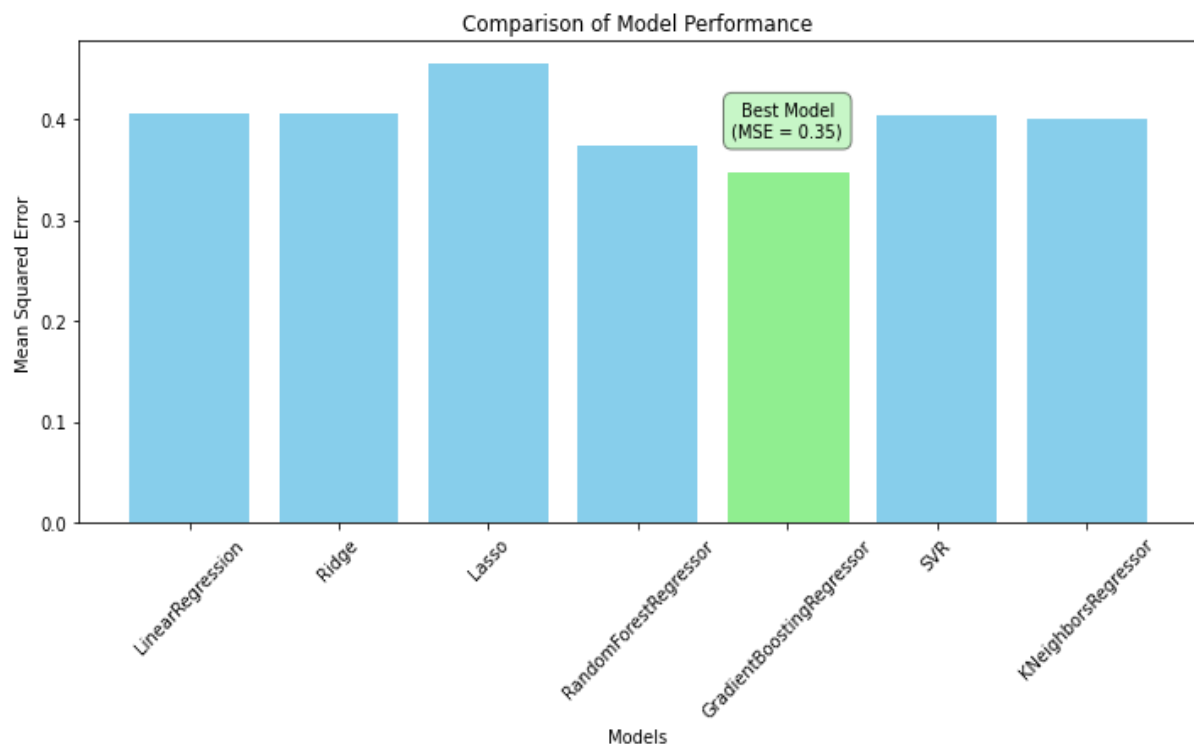
To sum up, the assessment of any model that has been put into practice is crucial to ascertain how well it predicts events linked to earthquakes. We were able to evaluate each model's performance in-depth by using suitable evaluation metrics, such as accuracy (for classification tasks), R-squared (R^2), or mean squared error (MSE).

The average squared difference between the predicted and actual values is measured by the mean squared error or MSE. Better performance is indicated by a reduced mean square error (MSE), which shows that the model's predictions are more accurate. On the other hand, R-squared (R^2) quantifies the percentage of the target variable's variance that the model can account for. A greater R-squared value indicates that more of the variability in the data is captured by the model. the data, suggesting increased capacity for prediction.

Accuracy is a frequently used metric to assess the model's performance in classification tasks. It shows the percentage of cases in the dataset that have been accurately classified out of all instances. Better performance in accurately predicting earthquake events or their categories is indicated by a higher accuracy.

We can determine which model is most appropriate for the given predicting task by evaluating each model's performance using these assessment metrics. After that, the chosen model can be used for real-world tasks including risk assessment, earthquake forecasting, and disaster preparedness planning.

To summarize, the assessment of model performance through suitable metrics is essential for well-informed decision-making and the effective utilization of predictive models in real-life situations.



CONCLUSION

In conclusion, the development of an earthquake prediction model using machine learning represents a significant step forward in mitigating the impact of seismic events on communities and infrastructure. Through meticulous data collection, cleaning, and preprocessing, coupled with sophisticated data visualization techniques, we gained invaluable insights into seismic patterns and characteristics. Feature selection allowed us to identify key predictors of earthquake occurrence, harnessing the power of machine learning algorithms to build a predictive model.

Throughout the project, we explored various machine learning techniques, from decision trees to deep learning models, striving to achieve the highest level of accuracy and reliability in earthquake prediction. Model training involved iterative refinement and validation, ensuring robustness and generalizability across different datasets and scenarios. By leveraging advanced methodologies and tools, we not only developed a predictive model but also contributed to the broader scientific understanding of earthquake dynamics.

In deploying the earthquake prediction model, we recognize the critical importance of real-time monitoring and continuous validation to enhance its effectiveness and responsiveness to evolving seismic conditions. While our model demonstrates promising accuracy, we acknowledge the ongoing challenges and uncertainties inherent in earthquake forecasting. Thus, we emphasize the need for collaboration and interdisciplinary research to advance the field further.

In summary, our earthquake prediction model serves as a testament to the potential of machine learning in addressing complex societal challenges. By harnessing data-driven insights and predictive analytics, we aim to empower stakeholders with timely information and actionable insights to enhance preparedness and resilience in the face of seismic events. Through continued innovation and collaboration, we remain committed to advancing the science of earthquake prediction for the benefit of society.

Future Work

The future work of a project focused on predictive modeling of earthquakes involves several avenues of research and development aimed at improving the accuracy and reliability of earthquake forecasting.

Some potential areas for future work in this project may include:

- Data Integration and Analysis
- Model Development and Refinement
- Integration of Multi-scale Modeling
- Uncertainty Quantification
- Real-time Monitoring and Early Warning Systems
- Validation and Verification
- Community Engagement and Stakeholder Collaboration

Advancing data integration and analysis methods will involve consolidating diverse datasets, including seismic records, geodetic measurements, and geological surveys, to build comprehensive models. Advanced analytical techniques such as machine learning algorithms will then be applied to identify intricate patterns and precursory signals associated with seismic activity.

Continuous model development and refinement efforts will focus on capturing the complex interactions between geological factors, fault dynamics, and tectonic processes influencing earthquake occurrence. This entails exploring various modeling approaches, including physics-based simulations and statistical models, to improve the accuracy and reliability of predictions.

The integration of multi-scale modeling techniques will be crucial for accounting for the hierarchical nature of seismic processes occurring across different spatial and temporal scales. By incorporating regional, fault-specific, and local-scale models, more precise and localized earthquake forecasts can be generated.

Moreover, enhancing real-time monitoring capabilities and early warning systems will be a priority. By integrating predictive models with advanced seismic monitoring networks and sensor technologies, timely detection of seismic precursors and issuance of early warnings can mitigate the impacts of earthquakes on at-risk populations.

Ultimately, stakeholder engagement and collaboration will be essential for the success of the project. By fostering partnerships with government agencies, academic institutions, and local communities, insights and expertise can be shared to co-develop tailored earthquake prediction solutions that address specific geographical regions' needs, enhancing overall disaster resilience.

Overall, the future work of a project focused on the predictive modeling of earthquakes involves a multidisciplinary approach integrating scientific research, technological innovation, and stakeholder engagement to advance our understanding of seismic processes and improve earthquake forecasting capabilities for enhanced risk management and disaster resilience.

REFERENCES

- [1] M. Akhoondzadeh and F. J. Chehrebargh. Feasibility of anomaly occurrence in aerosols time series obtained from modis satellite images during hazardous earthquakes. *Advances in Space Research*, 2016.
- [2] G. Asencio-Cortés, F. Martínez-Álvarez, A. Morales-Esteban, and J. Reyes. A sensitivity study of seismicity indicators in supervised learning to improve earthquake prediction. *Knowledge-Based Systems*, 101:15–30, 2016.
- [3] G. Asencio-Cortés, F. Martínez-Álvarez, A. Morales-Esteban, J. Reyes, and A. Troncoso. Improving earthquake prediction with principal component analysis: application to chile. In *International Conference on Hybrid Artificial Intelligence Systems*, pages 393–404. Springer, 2015.
- [4] A. Boucouvalas, M. Gkasios, N. Tselikas, and G. Drakatos. Modified- fibonacci-dual-lucas method for earthquake prediction. In *Third International Conference on Remote Sensing and Geoinformation of the Environment*, pages 95351A–95351A. International Society for Optics and Photonics, 2015.
- [5] J. Fan, Z. Chen, L. Yan, J. Gong, and D. Wang. Research on earthquake prediction from infrared cloud images. In *Ninth International Symposium on Multispectral Image Processing and Pattern Recognition (MIPPR2015)*, pages 98150E–98150E. International Society for Optics and Photonics, 2015.
- [6] E. Florido, F. Martínez-Álvarez, A. Morales-Esteban, J. Reyes, and J. Aznarte-Mellado. Detecting precursory patterns to enhance earthquake prediction in chile. *Computers & Geosciences*, 76:112–120, 2015.
- [7] A. Graves. Generating sequences with recurrent neural networks. *arXiv preprint arXiv:1308.0850*, 2013.
- [8] M. Hayakawa. Earthquake prediction with electromagnetic phenomena. In *THE IRAGO CONFERENCE 2015: 360 Degree Outlook on Critical Scientific and Technological Challenges for a Sustainable Society*, volume 1709, page 020002. AIP Publishing, 2016.
- [9] M. Hayakawa, H. Yamauchi, N. Ohtani, M. Ohta, S. Tosa, T. Asano, A. Schekotov, J. Izutsu, S. M. Potirakis, and K. Eftaxias. On the precursory abnormal animal behavior and electromagnetic effects for the kobe earthquake (m⁶) on april 12, 2013. *Open Journal of Earthquake Research*, 5(03):165, 2016.
- [10] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997.
- [11] M. Jiang. Easily magnetic anomalies earthquake prediction. In *MATEC Web of Conferences*, volume 63, page 01020. EDP Sciences, 2016.
- [12] S. Kannan. Innovative mathematical model for earthquake prediction. *Engineering Failure Analysis*, 41:89–95, 2014.
- [13] V. Korepanov. Possibility to detect earthquake precursors using cubesats. *Acta Astronautica*, 128:203–209, 2016.
- [14] M. Last, N. Rabinowitz, and G. Leonard. Predicting the maximum earthquake magnitude from seismic data in israel and its neighboring countries. *PloS one*, 11(1):e0146101, 2016.
- [15] C. Li and X. Liu. An improved pso-bp neural network and its application to earthquake prediction. In *Control and Decision Conference (CCDC), 2016 Chinese*, pages 3434–3438. IEEE, 2016.
- [16] J. Mahmoudi, M. A. Arjomand, M. Rezaei, and M. H. Mohammadi. Predicting the earthquake magnitude using the multilayer perceptron neural network with two hidden layers. *Civil Engineering Journal*, 2(1):1–12, 2016.
- [17] M. Moustira, M. Avraamides, and C. Christodoulou. Artificial neural networks for earthquake prediction using time series magnitude data or seismic electric signals. *Expert systems with applications*, 38(12):15032–15039, 2011.
- [18] S. Narayanakumar and K. Raja. A bp artificial neural network model for earthquake magnitude prediction in himalayas, india. *Circuits and Systems*, 7(11):3456, 2016.
- [19] A. Panakkat and H. Adeli. Recurrent neural network for approximate earthquake time

- and location prediction using multiple seismicity indicators. *Computer-Aided Civil and Infrastructure Engineering*, 24(4):280–292, 2009.
- [20] G. Pararas-Carayannis. The earthquake of May 12, 2008, in the Sichuan province of China. Website: <http://www.drgeorgepc.com/Earthquake2008ChinaSichuan.html>, 2008.
 - [21] S. Saba, F. Ahsan, and S. Mohsin. Bat-ann based earthquake prediction for Pakistan region. *Soft Computing*, pages 1–9, 2016.
 - [22] J. Shore and R. Johnson. Axiomatic derivation of the principle of maximum entropy and the principle of minimum cross-entropy. *IEEE Transactions on Information Theory*, 26(1):26–37, 1980.
 - [23] G. A. Sobolev. Methodology, results, and problems of forecasting earthquakes. *Herald of the Russian Academy of Sciences*, 85(2):107–111, 2015.
 - [24] N. Srivastava, G. E. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014.
 - [25] J. Thomas, F. Masci, and J. Love. A report that the 2012 m6: 0 earthquakes predicted after seeing an unusual cloud formation: Natural *Hazards and Earth System Sciences (NHESS)*, 2015.